A (Square $n \times n$)

$\underbrace{A\underline{x} = \lambda \underline{x}}$ (non-zero $\underline{x}$, numbers λ).

$$A x - \lambda I_n \underline{x} = \underline{0}$$

$$(A - \lambda I_n)\underline{x} = \underline{0} \checkmark$$

A has free cols iff $\det(A) = 0$.

If A has free cols; say $\underline{c}_j = \sum_{i<j} \alpha_i \underline{c}_i$

$$\det \begin{bmatrix} | & & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \underline{c}_j & \cdots & \underline{c}_n \\ | & & | & | & & | \end{bmatrix} = \det \begin{bmatrix} | & & | & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \alpha_1\underline{c}_1 + \alpha_2\underline{c}_2 + \cdots + \alpha_{j-1}\underline{c}_{j-1} & \underline{c}_{j+1} & \cdots & \underline{c}_n \\ | & & | & | & | & & | \end{bmatrix}$$

$$= \sum_{i<j} \alpha_i \det \begin{bmatrix} | & & | & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \underline{c}_i & \underline{c}_{j+1} & \cdots & \underline{c}_n \\ | & & | & | & | & & | \end{bmatrix} = 0.$$

If $\det(A) = 0$ then it has free cols.

OR If A has no free cols, then $\det(A) \neq 0$.

$\det(A) = \pm$ product of pivots. $\neq 0$.

$\pm$ depending on how many row exchanges in elimination.

<u>Check</u> $\det(A - \lambda I) = 0$.

$$A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}. \qquad A - \lambda I = \begin{pmatrix} 1-\lambda & 0 \\ -1 & 1-\lambda \end{pmatrix}.$$

$$\lambda I = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}.$$

$\det(A - \lambda I) = ?$ Use linearity properties.

First row $\underline{(1-\lambda)(1 \quad 0)} + 0(0 \quad 1)$

$$\det(A-\lambda I) = \det\begin{pmatrix} 1-\lambda\,[1 & 0] \\ -1 & (1-\lambda)\end{pmatrix} + \det\begin{pmatrix} 0\,[0 & 1] \\ -1 & 1-\lambda\end{pmatrix}.$$

$$= (1-\lambda)\underbrace{\det\begin{pmatrix} 1 & 0 \\ -1 & 1-\lambda\end{pmatrix}} + \cancel{0 \det}\begin{pmatrix} 0 & 1 \\ -1 & 1-\lambda\end{pmatrix}.$$

Second row: $-1\begin{bmatrix} 1 & 0\end{bmatrix} + (1-\lambda)\begin{bmatrix} 0, & 1\end{bmatrix}$

$$(1-\lambda)\det\begin{bmatrix} 1 & 0 \\ -1\,[1\;0] + (1-\lambda)[0\;1] & \end{bmatrix} = (1-\lambda)\det\begin{bmatrix} 1 & 0 \\ -1[1 & 0]\end{bmatrix} + (1-\lambda)\det\begin{bmatrix} 1 & 0 \\ (1-\lambda)[0. & 1]\end{bmatrix}$$

$$= (1-\lambda)(-1)\underbrace{\cancel{\det\begin{bmatrix} 1 & 0 \\ 1 & 0\end{bmatrix}}}_{0} + (1-\lambda)^2\underbrace{\det\begin{bmatrix} 1 & 0 \\ 0 & 1\end{bmatrix}}_{1}.$$

$$= \underbrace{(1-\lambda)^2}.$$

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1\end{bmatrix}. \qquad \det(A-\lambda I) = \underbrace{(1-\lambda)}^2.$$

roots of $\det(A-\lambda I)=0$ are $1, 1$.

$$A - 1I = \begin{bmatrix} 1 & 0 \\ -1 & 1\end{bmatrix} - 1\begin{bmatrix} 1 & 0 \\ 0 & 1\end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & 0\end{bmatrix}.$$

$$\text{null}(A-I) = \left\{\alpha\begin{pmatrix} 0 \\ 1\end{pmatrix} : \alpha \in \mathbb{R}\right\}.$$

$$\text{rref}(A-I) = \text{rref}\begin{bmatrix} 0 & 0 \\ -1 & 0\end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0\end{bmatrix}.$$

$\underline{c2} = 0\,\underline{c1}$

$\underline{c2} - \underline{0}\,\underline{c1} = \underline{0}$

Roots of $\underbrace{\det(A-\lambda I)=0}_{\text{characteristic polynomial.}}$: Eigen values.

If a root is repeated k times in $\det(A-\lambda I)=0$, we say that eigen value has <u>algebraic</u> multiplicity k.

Square Symmetric real A:

$A^T = A$ eg: $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

$$A - \lambda I = \begin{pmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{pmatrix}.$$

$\det(A - \lambda I) = (1-\lambda)^2 - 1 = \lambda^2 - 2\lambda + 1 - 1 = \lambda^2 - 2\lambda$.

Roots: 0, 2.

$\text{null}(A - 0I) = \text{null}(A) = \left\{\alpha\begin{pmatrix} 1 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R}\right\}$.

$\text{rref}(A) = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$. $\underline{c}2 = -\underline{c}1$

$\underline{c}2 + \underline{c}1 = \underline{0}$.

$\text{null}(A - 2I) = \text{null}\left[\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}\right] = \left\{\alpha\begin{pmatrix} 1 \\ -1 \end{pmatrix} : \alpha \in \mathbb{R}\right\}$.

Dot product of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0$.

The eigenvectors (one from each of null(A) and null$(A-2I)$)
$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ form a max linearly indpt set in $\mathbb{R}^2$.

For all $\underline{x} \in \mathbb{R}^2$, $\underline{x} = \alpha\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 2\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$\begin{aligned} A\underline{x} &= A\left[\alpha\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right] \\ &= A\left[\alpha\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right] + A\left[\beta\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right] \\ &= \alpha A\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \\ &= \alpha . 0 . \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2\beta\begin{pmatrix} 1 \\ -1 \end{pmatrix}. \end{aligned}$$

$$A(A\underline{x}) = 4\beta\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$B = \begin{pmatrix}1\\-1\end{pmatrix}\begin{pmatrix}1 & -1\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}1 & 1\end{pmatrix} = \begin{pmatrix}1 & -1\\-1 & 1\end{pmatrix} + \begin{pmatrix}1/2 & 1/2\\1/2 & 1/2\end{pmatrix}$$

$$= \begin{pmatrix}3/2 & 1/2\\1/2 & 3/2\end{pmatrix}.$$

$$\lambda I = \lambda\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} = \begin{pmatrix}\lambda & 0\\0 & \lambda\end{pmatrix}$$

Eigenvalues of B.

$$\det(B - \lambda I) = \det\begin{pmatrix}3/2-\lambda & 1/2\\1/2 & 3/2-\lambda\end{pmatrix}.$$

$$\begin{bmatrix}\frac{3}{2}-\lambda & 1/2\\1/2 & 3/2-\lambda\end{bmatrix} \xrightarrow[\ ]{r2 \leftarrow r2 - \frac{\frac{1}{2}}{3/2-\lambda} r1} \begin{bmatrix}3/2-\lambda & 1/2.\\0 & \left(\frac{3}{2}-\lambda\right) - \frac{\frac{1}{2}\cdot\frac{1}{2}}{\left(\frac{3}{2}-\lambda\right)}.\end{bmatrix} \quad \lambda \neq 3/2.$$

$$\det(B) = (3/2-\lambda)\left(\frac{3}{2}-\lambda\right) - 1/4. = \lambda^2 - 3\lambda + 9/4 - 1/4 = \lambda^2 - 3\lambda + 2.$$
$$= (\lambda-2)(\lambda-1)$$

Roots : 2, 1

$\lambda = 2$

$$B - \lambda I = \begin{pmatrix}\frac{3}{2}-\lambda & 1/2\\1/2 & 3/2-\lambda\end{pmatrix} = \begin{pmatrix}-1/2 & 1/2\\1/2 & -1/2\end{pmatrix}$$

$\underline{c2} = -c1.$

$\underline{c}1 + \underline{c}2 = 0.$

$\therefore$ Eigenvector for eigenvalue $\lambda = 2$ $\underline{\begin{pmatrix}1\\1\end{pmatrix}}.$

$$\lambda = 1 \quad B - \lambda I = \begin{pmatrix}1/2 & 1/2\\1/2 & 1/2\end{pmatrix}$$

$\underline{c\,2} = \underline{c1}.$

Eigenvector for eigenvalue $\underline{\underline{\lambda = 1}}$ $\begin{pmatrix}1\\-1\end{pmatrix}.$

$$\underline{x} = \begin{pmatrix}1\\0\end{pmatrix}. = \frac{1}{2}\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1\\-1\end{pmatrix}.$$

$$B\underline{x} = \frac{1}{2}B\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}B\begin{pmatrix}1\\-1\end{pmatrix} = \frac{1}{2}\cdot 2\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}\cdot 1\cdot\begin{pmatrix}1\\-1\end{pmatrix}.$$

$\begin{pmatrix}1\\1\end{pmatrix}$ is in null $(B - 2I)$ $\quad (B-2I)\begin{pmatrix}1\\1\end{pmatrix} = 0. \quad B\begin{pmatrix}1\\1\end{pmatrix} - 2\begin{pmatrix}1\\1\end{pmatrix} = 0.$

$$B\begin{pmatrix}1\\1\end{pmatrix} = 2\begin{pmatrix}1\\1\end{pmatrix}$$

$$B\begin{pmatrix}1\\-1\end{pmatrix} = 1\cdot\begin{pmatrix}1\\-1\end{pmatrix}$$

$$B(B\underline{x}) = B\left[\tfrac{1}{2}\cdot 2\begin{pmatrix}1\\1\end{pmatrix} + \tfrac{1}{2}\cdot 1\begin{pmatrix}1\\-1\end{pmatrix}\right] = \tfrac{1}{2}\cdot 2^2\begin{pmatrix}1\\1\end{pmatrix} + \tfrac{1}{2}\cdot 1^2\begin{pmatrix}1\\-1\end{pmatrix}.$$

$$B^n(\underline{x}) = \tfrac{1}{2}\cdot 2^n\begin{pmatrix}1\\1\end{pmatrix} + \tfrac{1}{2}\cdot (1)^n\begin{pmatrix}1\\-1\end{pmatrix}.$$

Spectral Decomposition Theorem.

Suppose A is square, symmetric. $n\times n$ matrix.

(i) A has n real eigenvalues (possibly repeated).

$\det(A-\lambda I)$ has n real roots (possibly repeated).

(ii) Eigenvectors corresponding to distinct eigenvalues are orthogonal.

(iii) Suppose λ^* is an eigenvalue $(A-\lambda^* I)$. has free columns.

$\dim[\text{null}(A-\lambda^* I)]$ = # free cols in $A-\lambda^* I$

= # λ^* is repeated among roots of $\det(A-\lambda I)$

$\Rightarrow$ we get n linearly indpt eigenvectors.

$\therefore$ eigenvectors form a basis for $\mathbb{R}^n$.

we can also get n orthogonal vectors as a basis for $\mathbb{R}^n$.

Normalize all so that they have length 1.

$\underline{v}_1 \ldots \underline{v}_n$.

$$\begin{cases} A\underline{v}_1 = \lambda_1 \underline{v}_1, \\ A\underline{v}_2 = \lambda_2 \underline{v}_2 \\ \vdots \\ A\underline{v}_n = \lambda_n \underline{v}_n. \end{cases}$$

$$\underline{v}_i^T\underline{v}_j = \begin{cases} 0 & \text{if } i\neq j \\ 1 & \text{if } i=j. \end{cases}$$

$$V = \begin{bmatrix} | & & | \\ \underline{v}_1 & \cdots & \underline{v}_n \\ | & & | \end{bmatrix}.$$

$$V^TV = \begin{bmatrix} ij^{\text{th}} \text{ entry} \\ = \underline{v}_i^T\underline{v}_j \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$$\underbrace{V^{-1} = V^T}.$$

$$AV = A\begin{bmatrix} \underline{v}_1 & \cdots & \underline{v}_n \end{bmatrix} = \begin{bmatrix} \lambda_1 \underline{v}_1 & \lambda_2 \underline{v}_2 & \cdots & \lambda_n \underline{v}_n \end{bmatrix}$$

$$= V \underbrace{\begin{bmatrix} \lambda_1 & 0 & & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & \vdots & & \vdots \\ 0 & 0 & & \lambda_n \end{bmatrix}}_{\Lambda}.$$

$$= V\Lambda.$$

$$A \underbrace{VV^T}_{I} = V\Lambda V^T \qquad \therefore \quad A = V\Lambda V^T$$

$\det(A - \lambda I) = 0.$

Roots $\quad \lambda_1 \geq \lambda_2 \geq \lambda_3 \cdots \geq \lambda_n.$

Distinct roots
λ_{i_1} repeats n_1 times.
λ_{i_2} repeats n_2 times.
$\vdots$
λ_{i_k} —"— n_k times.

Clearly $n_1 + n_2 + \cdots n_k = n.$

$i_1 = 1 \quad i_2 = 2 \quad i_3 = 4 \quad i_4 = 5$

$\lambda_1 > \lambda_2 = \lambda_3 > \lambda_4 > \lambda_5 = \lambda_6.$

$1, \ 0.5, \ 0.5, \ 0.2, \ -0.5, -0.5$

λ_1 has multiplicity 1.
λ_2 has —"— 2.
λ_4 has —"— 1.
λ_5 has —"— 2.

$1 + 2 + 1 + 2 = 6.$

null $(A - \lambda_{i_1} I)$ contributes n_1 linearly independent vectors.
null $(A - \lambda_{i_2} I)$ —"— n_2 —"— —"—
$\vdots$
null $(A - \lambda_{i_k} I)$ —"— n_k —"— —"—.

Concatenate these lists (because vectors in diff eigenspaces above are going to be orthogonal if A is symmetric) to form a list of $n_1 + n_2 + \cdots + n_k = n$ linearly indpt vectors in $\mathbb{R}^n$.

Any vector in $\mathbb{R}^n$ is a linear combination of the above.
Let $\underline{v}_1 \ldots \underline{v}_n$ be n mutually orthogonal (unit-length) vectors.

$A = V\Lambda V^T \qquad V = \begin{bmatrix} \underline{v}_1 \cdots & \underline{v}_n \end{bmatrix}.$

$$\Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}.$$

$$A V = V \Lambda.$$

$$V^T A V = \underbrace{(V^T V)}_{=I} \Lambda = \Lambda. = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

* Interpret $A\underline{x}$ for any $\underline{x} \in \mathbb{R}^n$.

$\underline{x} = \alpha_1 \underline{v}_1 + \dots + \alpha_n \underline{v}_n$ (because $\underline{v}_1 \dots \underline{v}_n$ is a basis).

$A\underline{x} = \alpha_1 A\underline{v}_1 + \alpha_2 A\underline{v}_2 + \dots + \alpha_n A\underline{v}_n.$

$= \alpha_1 \lambda_1 \underline{v}_1 + \alpha_2 \lambda_2 \underline{v}_2 + \dots + \alpha_n \lambda_n \underline{v}_n$

(for symmetric matrices).

$(A\underline{x})^T = x^T A^T$

* Suppose X : $m \times n$ matrix. $(AB)^T = B^T A^T.$

$(X^T X)^T = X^T (X^T)^T = X^T X.$ $(n \times n)$

$(X X^T)^T = (X^T)^T X^T = X X^T.$ $(m \times m).$

$m \geq n.$ (X is tall).

Eigenvalues of $X^T X$ are $\lambda_1 \dots \lambda_n$. ≥ 0.

If $\lambda_n < 0$. $(X^T X) \underline{v}_n = \lambda_n \underline{v}_n.$

$v_n^T (X^T X) \underline{v}_n = \lambda_n v_n^T v_n = \lambda_n$ (we chose $\underline{v}_n$ that way).

$(X\underline{v}_n)^T X\underline{v}_n = (\underline{v}_n^T X^T)(X\underline{v}_n).$

$\|\underline{w}_n\|^2 = w_n^T \underline{w}_n.$

Positive semi-definite.

* Eigenvalues of $X X^T$: Every e.v. of $X^T X$ is an e.v. of $X X^T$.

Let $\lambda_1, \dots \lambda_n$ be e.v of $X^T X$.

$X X^T X \underline{v}_i = \lambda_i X \underline{v}_i.$ null (X) = null $(X^T X)$.

$(X X^T) X \underline{v}_i = \lambda_i X \underline{v}_i.$ if $X^T X \underline{v}_i \neq 0$.

if $\underbrace{X^T X \underline{v}_i \neq 0}_{\lambda_i \neq 0}$, then $X \underline{v}_i \neq 0$.

means $\lambda_i > 0$. (because all e.v. of X^TX are ≥ 0).

If $\lambda_i \neq 0$, then let $w_i = X\underline{v}_i$ $\quad w_i \neq 0$.

we have $XX^T\underline{w}_i = \lambda_i \underline{w}_i$

So w_i is an eigenvector of XX^T with eigenvalue λ_i.

Suppose $\text{rank}(X) = k$. $(\leq \min(m,n))$.

$\hookrightarrow$ # pivots in X.

free cols in X : $n-k$. $\therefore$ $\dim(\text{null}(X)) = n-k$.

$\text{null}(X) = \text{null}(X^TX)$ $\therefore$ $\dim(\text{null}(X^TX)) = n-k$.

multiplicity of eigenvalue 0 in X^TX $= n-k$.

XX^T: $\quad \text{null}(X^T) = \text{null}((X^T)^T X^T) = \text{null}(XX^T)$.

X^T also has k pivots.

free cols in X^T : $m-k = (m-n) + n-k$.

XX^T has $m-k$ free cols as well.

multiplicity of eigenvalue 0 in XX^T $= m-k = (m-n) + (n-k)$.

$\therefore$ XX^T and X^TX share the same non-zero eigenvalues.

if $\lambda_i \neq 0$. $\underline{v}_i$ is an eigenvector of X^TX.

$X^TX\underline{v}_i = \lambda_i \underline{v}_i$

$\therefore$ $XX^TX\underline{v}_i = \lambda_i X\underline{v}_i$ $\quad$ Set $w_i = X\underline{v}_i$

$XX^T\underline{w}_i = \lambda_i \underline{w}_i$

If $\lambda_i \neq 0$ and $\underline{w}_i$ is an eigenvector of XX^T.

$XX^T\underline{w}_i = \lambda_i \underline{w}_i$

$X^TXX^T\underline{w}_i = \lambda_i X^T\underline{w}_i$ $\quad$ Set $\underline{v}_i = X^T\underline{w}_i$

$X^TX\underline{v}_i = \lambda_i \underline{v}_i$.

$$\left.\begin{array}{l}\text{Suppose } \underline{u} \text{ is a unit vector (vector with length 1).}\\ \quad\text{Find } \underline{u} \text{ such that } \|X\underline{u}\| \text{ is maximized.}\end{array}\right\}.$$

$\|X(2\underline{u})\| = 2\|X\underline{u}\|$ $\times$

For any vector $\|\underline{z}\|^2 = \underline{z}^T \underline{z}$

$(\underline{u}^T A \underline{u})$

$\|X\underline{u}\|^2 = (X\underline{u})^T(X\underline{u}) = \underline{u}^T X^T X \underline{u}$.

Suppose eigenvalues of $X^T X$ are $\lambda_1 \geq \lambda_2 \ldots \geq \lambda_n$.

Suppose $\underline{v}_1 \ldots \underline{v}_n$ are an orthonormal eigenvector basis for $\mathbb{R}^n$.

$$\underline{v}_i^T \underline{v}_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j. \end{cases}$$

$\underline{u} = \alpha_1 \underline{v}_1 + \ldots + \alpha_n \underline{v}_n$.

$$\begin{aligned} X^T X \underline{u} &= X^T X(\alpha_1 \underline{v}_1 + \ldots + \alpha_n \underline{v}_n). \\ &= \alpha_1 X^T X \underline{v}_1 + \ldots + \alpha_n X^T X \underline{v}_n. \\ &= \alpha_1 \lambda_1 \underline{v}_1 + \ldots + \alpha_n \lambda_n \underline{v}_n. \end{aligned}$$

$$\begin{cases} A(2\underline{v}_1 + 3\underline{v}_2) \\ = A(2\underline{v}_1) + A(3\underline{v}_2) \\ = 2A\underline{v}_1 + 3A\underline{v}_2. \end{cases}$$

$$\begin{aligned} u^T(X^T X \underline{u}) &= (\alpha_1 \underline{v}_1^T + \ldots \alpha_n \underline{v}_n^T).(\alpha_1 \lambda_1 \underline{v}_1 + \ldots + \alpha_n \lambda_n \underline{v}_n). \\ &= \underline{\alpha_1^2 \lambda_1 + \alpha_2^2 \lambda_2 + \alpha_3^2 \lambda_3 + \ldots + \alpha_n^2 \lambda_n.} \end{aligned}$$

To maximize set $\alpha_1 = 1$. $\underline{u} = \underline{v}_1$.

rank$(X) = k$.

$$\underbrace{\lambda_1 \ldots \lambda_k}_{>0} \; \underbrace{\lambda_{k+1} \ldots \lambda_n}_{=0} \; \underbrace{\lambda_{n+1} \qquad \lambda_m}_{=0.}$$

$n \times n$	$X^T X$	$\underline{v}_1 \ldots \underline{v}_k \; \underline{v}_{k+1} \; \underline{v}_n$.	} orthonormal basis for $\mathbb{R}^n$
$m \times m$	$X X^T$	$\underline{w}_1 \quad \underline{w}_k \; \underline{w}_{k+1} \; \underline{w}_n \quad \underline{w}_{n+1} \ldots \underline{w}_m$.	} orthonorm basis for $\mathbb{R}^m$

$\underline{v}_1 \ldots \underline{v}_k$ are eigenvectors of $X^T X$ for non-zero eigenvalues $\lambda_1 \ldots \lambda_k$

$X\, X^T X \underline{v}_1 = X(\lambda_1 \underline{v}_1)$

$X X^T (X\underline{v}_1) = \lambda_1 (X\underline{v}_1). \Rightarrow X\underline{v}_1$ is an eigenvector for XX^T with e.v. λ_1.

1. $\|X\underline{u}\|$ is maximized when we set $\underline{u} = \underline{v}_1$.

$X\underline{v}_1$ has to be in the direction of $\underline{w}_1$

$$\underline{w}_1 = \frac{X\underline{v}_1}{\|X\underline{v}_1\|}$$

2. Let $\underline{u}$ be a unit vector in $\mathbb{R}^m$. What should be $\underline{u}$ so that $\|X^T \underline{u}\|$ is maximized.

Ans: $\underline{w}_1$. $X^T \underline{w}_1$ is in the direction of $\underline{v}_1$.

3. Think of all vectors $\perp \underline{v}_1$

Among all unit vectors $\perp \underline{v}_1$, which unit vector $\underline{u}$ maximizes $\|X\underline{u}\|$

Ans: $\underline{v}_2$.

$X\underline{v}_2$ is in the direction of $\underline{w}_2$.

$$X^T X = V \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} V^T. \qquad V = [\underline{v}_1 \; \cdots \; \underline{v}_n].$$

$$X X^T = W \begin{bmatrix} \lambda_1 & & & & 0 \\ & \ddots & & & \\ & & \lambda_n & & \\ & 0 & & 0 & \\ & & & & \ddots \; 0 \end{bmatrix} W^T. \qquad W = [\underline{w}_1 \; \cdots \; \underline{w}_m].$$

(X has rank k).

$$\underline{w}_1 = \frac{X\underline{v}_1}{\|X\underline{v}_1\|} \qquad \underline{w}_2 = \frac{X\underline{v}_2}{\|X\underline{v}_2\|} \quad \cdots \quad \underline{w}_k = \frac{X\underline{v}_k}{\|X\underline{v}_k\|}.$$

$$\underline{v}_1 = \frac{X^T\underline{w}_1}{\|X^T\underline{w}_1\|} \qquad \underline{v}_2 = \frac{X^T\underline{w}_2}{\|X^T\underline{w}_2\|} \quad \cdots \quad \underline{v}_k = \frac{X^T\underline{w}_k}{\|X^T\underline{w}_k\|}.$$

singular values.

$$X = W \begin{bmatrix} \sqrt{\lambda_1} & & 0 \\ 0 & \ddots & \sqrt{\lambda_n} \\ \hline & 0 & \end{bmatrix} V^T. \qquad \text{SVD of } X.$$

basis for cols — basis for rows.